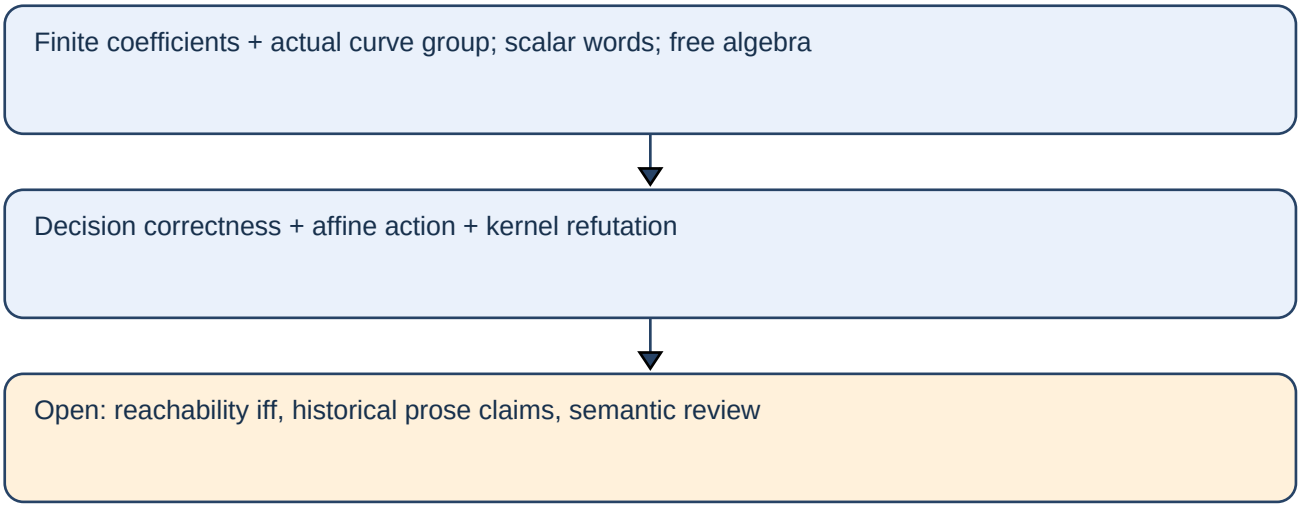
crypto-autoresearcher: checked statements and open bridges



Logical obligations only. No curve route or measured performance is represented.

Lean retrospective, 2026-10-18

Status: **partial retrospective**. This batch supplies the decomposition certificate target, the word-action affine-normal-form target, and the commutator-ideal refutation target. It also rechecks existing frozen formal targets. Independent semantic-fidelity review and the full historical review remain open; no official hypothesis or ledger status is changed.

Request and baseline

The owner requested retroactive review and Lean proofs of past claims in `aburan28/crypto`, `aburan28/cryptanalysis`, and `aburan28/crypto-autoresearcher`. The autoresearcher baseline is `6edd9e1ea14450fcb5a168137407ecf303e48773`. inventory.json records all 66 frozen formal targets and the 47 tracked Lean source paths excluding self-test fixtures at that revision. Four target files initially lacked source. This is a formal-source inventory, not an extraction of every claim in the research archive or earlier Git history.

Requirement	Status and evidence
Exact Git-tree inventory of existing formal targets	inventory.json, reproducible with `inventory.mjs`
Recheck declared targets	Per-target receipts; final current-source status in verification-summary.json
Fill the certificate target	Certificate.lean
Fill the affine-normal-form target	AffineNormalForm.lean
Fill the commutator-ideal refutation	CommutatorIdeal.lean
Exact shadow reachability iff	Open: `formal/targets/ncp-reachability.yaml`; the matching construction and length lower bound are not yet formalized
Full prose-claim inventory and independent semantic review	Open: coverage problem

Statements and readable proofs

The proof map gives exact declarations, source revisions, hypotheses, and verification evidence. The frozen target specs are preserved.

Semaev Lemma 3, from `THM_SEMAEV_FALL1.md`: the coefficient field k is finite, K is a decidable field extension, and the declared finite family spans the subspace. Membership is decided by enumerating the finite coefficient tuples. `inDeclaredSpan_iff` proves that this predicate is exactly Mathlib linear-span membership. A certificate is an arbitrary finite tuple of checked nonsingular affine points on an arbitrary Weierstrass curve. The checker tests the explicit curve equation, every x-coordinate's subspace membership, and the sum using Mathlib's actual elliptic-curve group law.

`checkDecomposition_iff` proves soundness and completeness for this predicate. The nonsingularity proofs are checked inputs to point construction; a parser for untrusted serialized coordinates is not supplied.

The computational instance is `instDecidableValidDecomposition`. The frozen target name `decidable_validDecomposition` names a theorem supplying a Boolean decision and its correctness contract. This distinction matters because the existing verifier requires a theorem declaration, while `Decidable` instances are data. The initial instance-only attempt and a subsequent elaboration failure are preserved as failed receipts, followed by the corrected recheck. The verifier was not relaxed. No relation-existence, solving-degree, complexity, or attack-success claim follows from certificate decidability.

NCP Lemma 1, from `THM_COMMUTATOR_KERNEL1.md`: letters execute left to right. Each translation occurrence is assigned the sign of its suffix; `wordSign_eq_negOne_pow` proves this is minus one to the number of negations to its right. Induction on a word proves `action(w,x)=sign(w)*x+sum(count_i(w)*s_i)` over every commutative ring. Consequently equal shadows give equal actions, and evaluation at 1 gives the source's orbit formula in ZMod n. The formal action uses the cyclic scalar model; concrete curve-point implementations and the source's stronger kernel and cost claims are not certified by this lemma.

NCP Proposition 1: in the genuine free associative integer algebra, v squared minus 1 maps to zero under translation/negation evaluation into the group algebra of the affine permutation group of ZMod n. It is outside the two-sided ideal of all commutators: the algebra homomorphism sending every free generator to the integer 2 kills commutators but sends this witness to 3. Therefore the two ideals are unequal. This is the main refutation, not the source's full presentation of the evaluation kernel. No surjectivity hypothesis is needed. Companion witnesses involving v*t_i, and a complexity-level simulation theorem, are not proved here.

Verification and retained failures

The existing Lean `v4.35.0-rc1` and Mathlib `7fa22735254ddf7e11f119250a9f2d0352bc7c3a` pins are unchanged. The scoped torsion workspace retains its separate Lean `v4.22.0` pin. See each receipt for the exact source/spec/toolchain/manifest hashes, build log, namespace audit and requested-declaration audit. The final main-workspace audit covers 498 theorem declarations using standard foundations only. Semantic review remains explicitly pending in every verifier receipt.

```
python tools/verify_formal_targets.py --artifact-dir /tmp/fresh-retrospective-receipts
```

Run from the repository root with `lake` and all pinned toolchains available. Use a fresh receipt directory; never overwrite the historical attempts here. The initial broad-run transcript is verify-all.log.gz (read with `gzip -dc`); later targeted rechecks resolve the documented instance/theorem mismatch and missing toolchain without rewriting that transcript. summarize.mjs accepts a successful receipt only if its theorem, spec, toolchain and manifest hashes match the current files.

The existing `formal-workspace` CI workflow discovers the added target files. `lean4checker` is not installed; fresh external rechecking was not performed.

Cross-repository coverage

Crypto's new batch proves conditional subgroup DLP transport on actual point groups and factor-base range containment; cryptanalysis's batch proves both rho collision formulas and a zero-denominator counterexample. Their separate reports identify assumptions and implementation gaps. A complete historical claim extraction, all remaining crypto-project builds, concrete geometric bridges, empirical replay and independent semantic review remain outstanding. The problem record keeps the original all-repositories requirement open.

Diagram, editable data, and report PDF illustrate the proof boundary. The reviewed theory notes' measured tables and the README's performance descriptions receive no new measurements; no performance graph, curve identity, or isogeny-route edge is changed by these proof additions. Historical scientific state is preserved.